

SHUBHI DIXIT

Aspiring Software Engineer | Second-Year B.Tech CSE Student | DSA • Web Dev • Machine Learning • AI-ML

[Gmail](#) • [LinkedIn](#) • [GitHub](#) • [Codeforces](#) • [LeetCode](#)

SUMMARY

Computer Science undergraduate at Delhi Technological University (DTU) with a strong foundation in DSA, full-stack development, AI, and distributed systems. Interested in building practical and scalable technology solutions to real-world problems.

EDUCATION

B.Tech (Computer Science Engineering) – Delhi Technological University (DTU)	2025–2029
CGPA: 6.5	
Class XII (CBSE) – 89.8%	2024–2025
Class X (CBSE) – 95.8%	2021–2022

SKILLS

Languages	C++, Python, JavaScript, TypeScript
Technical	Data Structures & Algorithms, Full-Stack Development, Machine Learning, Operating Systems, LLMs & RAG, OOP
Tools	React.js, Next.js, Node.js, Express.js, MongoDB, SQLite, ChromaDB, Weaviate, Git & GitHub

PROJECTS

MuskMelon – Digital Twin of Elon Musk | *Next.js, TypeScript, Temporal RAG, Weaviate Cloud, OpenAI* | [GitHub](#)

- Built an **adaptive, two-way, version-controlled knowledge twin**, transforming dated public sources into traceable **Knowledge Commits** with Now Mode, Time Lens, and Belief Diff for current, historical, and evolving positions.
- Designed a reusable **Context Capsule** to retain relevant chunks across follow-up queries and a consent-based **User Twin** to personalize response depth and tone without altering factual evidence.
- Orchestrated Google Drive → Weaviate → LLM execution through **Swytchcode**, integrating policy guardrails, Answer Receipts, and a Knowledge-Voice Firewall.

JeevanMesh – Self-Healing Disaster-Response Drone Swarm | *Next.js, TypeScript, Google Maps API, Distributed Systems, Multi-Agent Simulation* | [GitHub](#)

- Built a **leaderless, partition-tolerant drone-swarm simulation** that autonomously switches scout, verifier, relay and supply roles, transfers low-battery tasks, and continues operating from its last verified mission state during network failure.
- Designed **BubbleNet Adaptive Search** and **Firefly Relay Positioning** for confidence-driven survivor discovery and dynamic mesh repair, reducing target-discovery time by **39.7% over grid scanning**.
- Developed a real-time **disaster command centre** with drone telemetry and 3D positions, survivor uncertainty maps, blocked-route rerouting, store-carry-forward synchronisation, civilian JeevanLink connectivity, and human approval checkpoints for high-risk rescue actions.

ChronoSync | *ReactJS, Node.js, Express.js, MongoDB, Genetic Algorithm, Greedy Algorithm* | (SIH 2025) | [GitHub](#)

- Built a **web-based timetable scheduling system** for Indian universities, supporting multi-department scheduling, shared classroom/lab allocation, and faculty preference management.
- Designed a **Genetic Algorithm + Greedy Optimization** framework for hard/soft constraint satisfaction, enabling automated timetable generation, conflict detection, schedule regeneration and ERP integration.

NyayaBot | *Python, TypeScript, React, Gemma 4, Ollama, ChromaDB, SQLite* | [GitHub](#)

- Built a **100% on-device, offline legal-assistance engine** on Gemma 4, using Per-Layer Embeddings for inference under 6GB VRAM, with a **hierarchical ChromaDB RAG pipeline** and ShieldAI guardrails for Act→Section retrieval, citation verification, and safe outputs.
- Engineered a **ReAct-loop workflow orchestrator** coordinating 8 tools over a 13-table SQLite schema, using **SQL-level compare-and-swap concurrency** and idempotency-key validation for atomic, conflict-safe case operations.

Oon Nirnay — Evidence-first AI decision navigator for rural wool entrepreneurs using LLMs and prompt engineering. [Link](#)

SkillForge — Modular agentic AI runtime with Planning, Vision, Memory, and Execution agents. [GitHub](#)

RANKS AND ACHIEVEMENTS

JEE Main: 99.08 percentile | Qualified JEE Advanced

1st Place, *Guessapalooza* | IEEE DTU INVICTUS Annual Technical Fest | ₹3,000 Cash Prize

4th Place, VibeWright Hackathon | Oblivion'26, NSUT | Among 250+ teams

2nd Place, State-Level Mental Mathematics Competition (Class XII) | ₹10,500 Cash Prize

Rank 12, MVPP Delhi State Scholarship Examination (Class IX), among 20,000+ students | ₹5,000 Scholarship

Solved 200+ DSA problems

EXTRA-CURRICULAR ACTIVITIES

- Volunteered with NSS, mentoring underprivileged children and supporting educational outreach; also participated in debate, poetry, and public-speaking activities throughout school.
- Completed a 2-month Summer Program at the British Council, enhancing spoken English, creative writing, presentation and leadership skills.